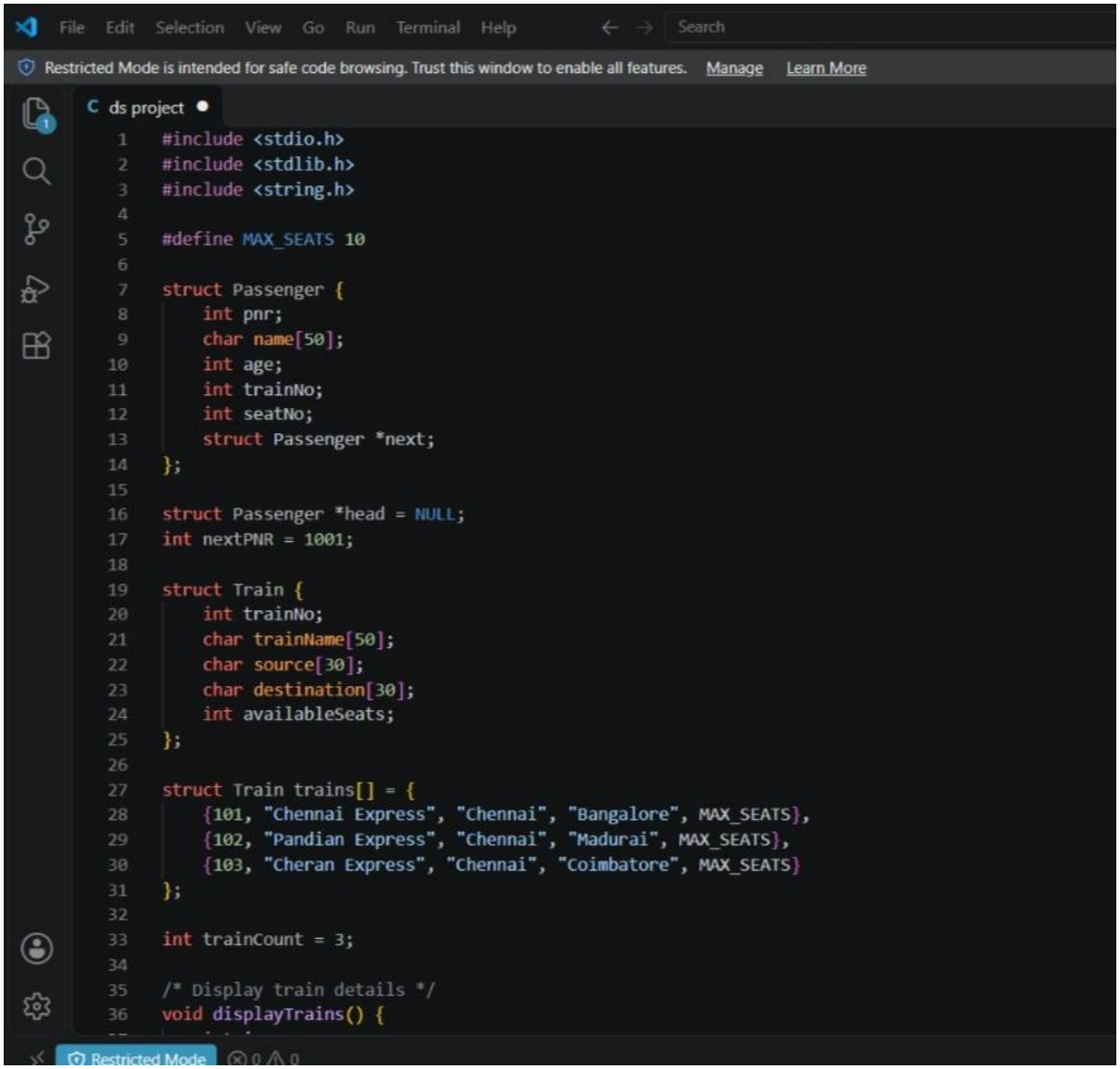
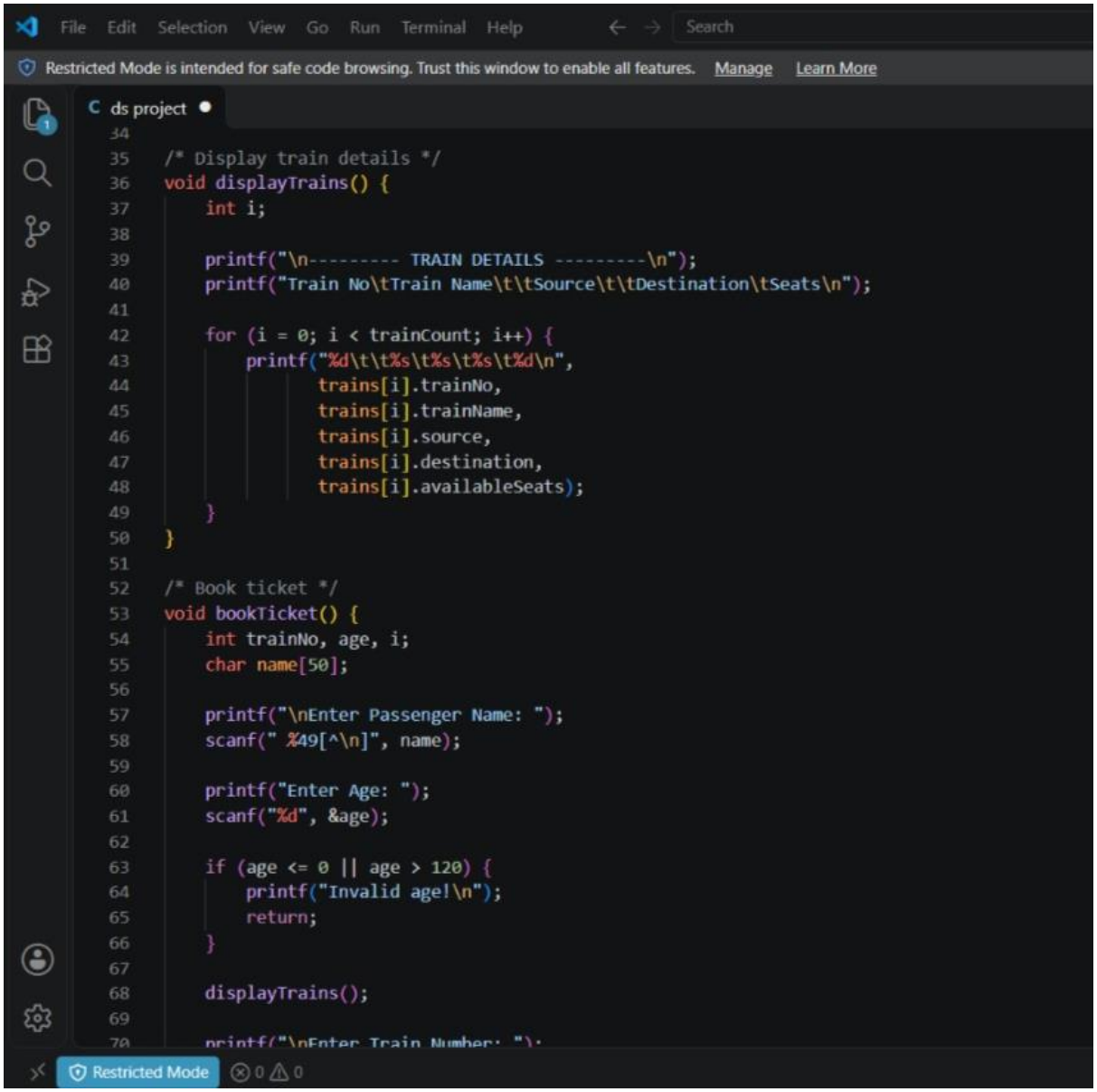
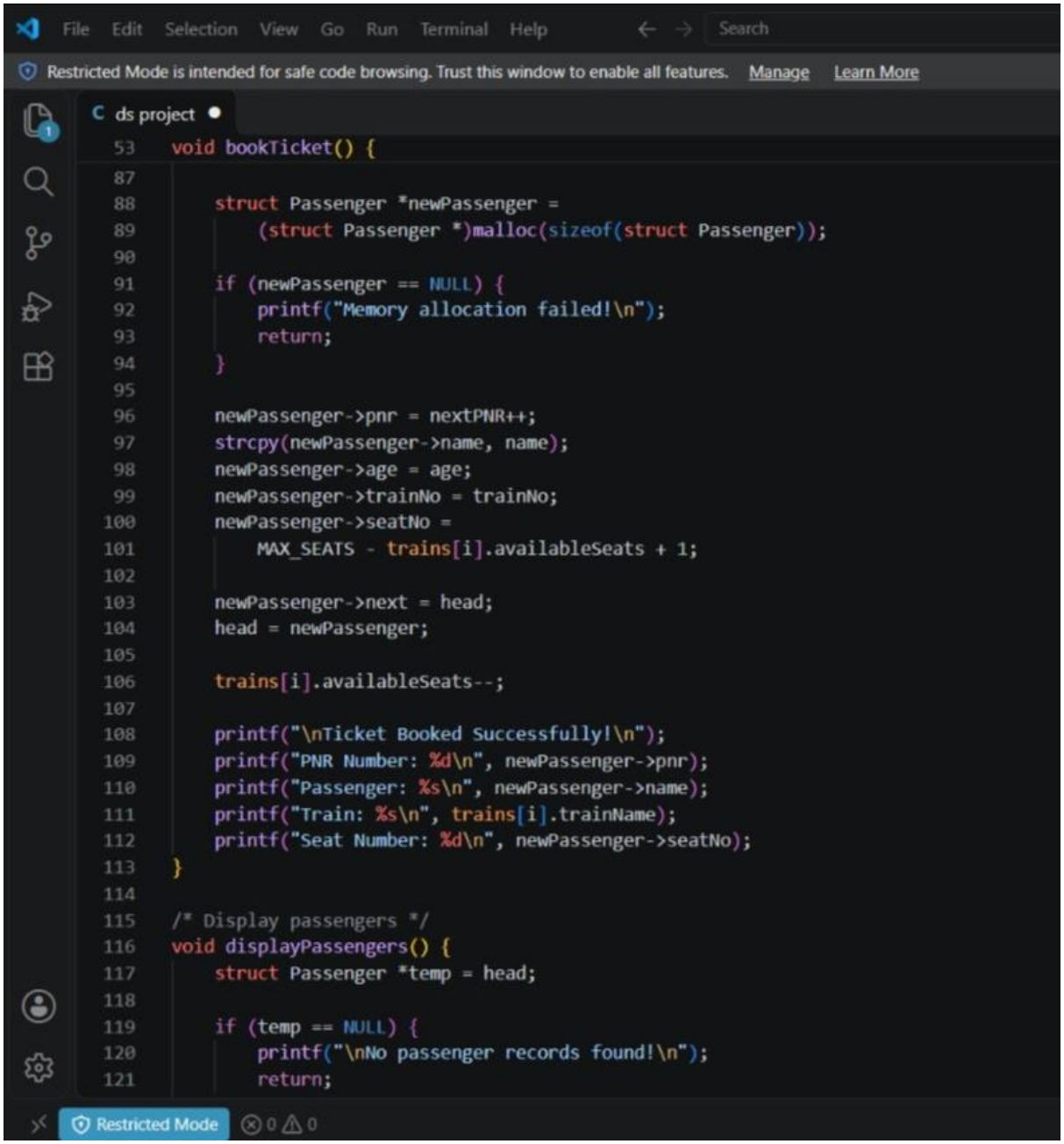
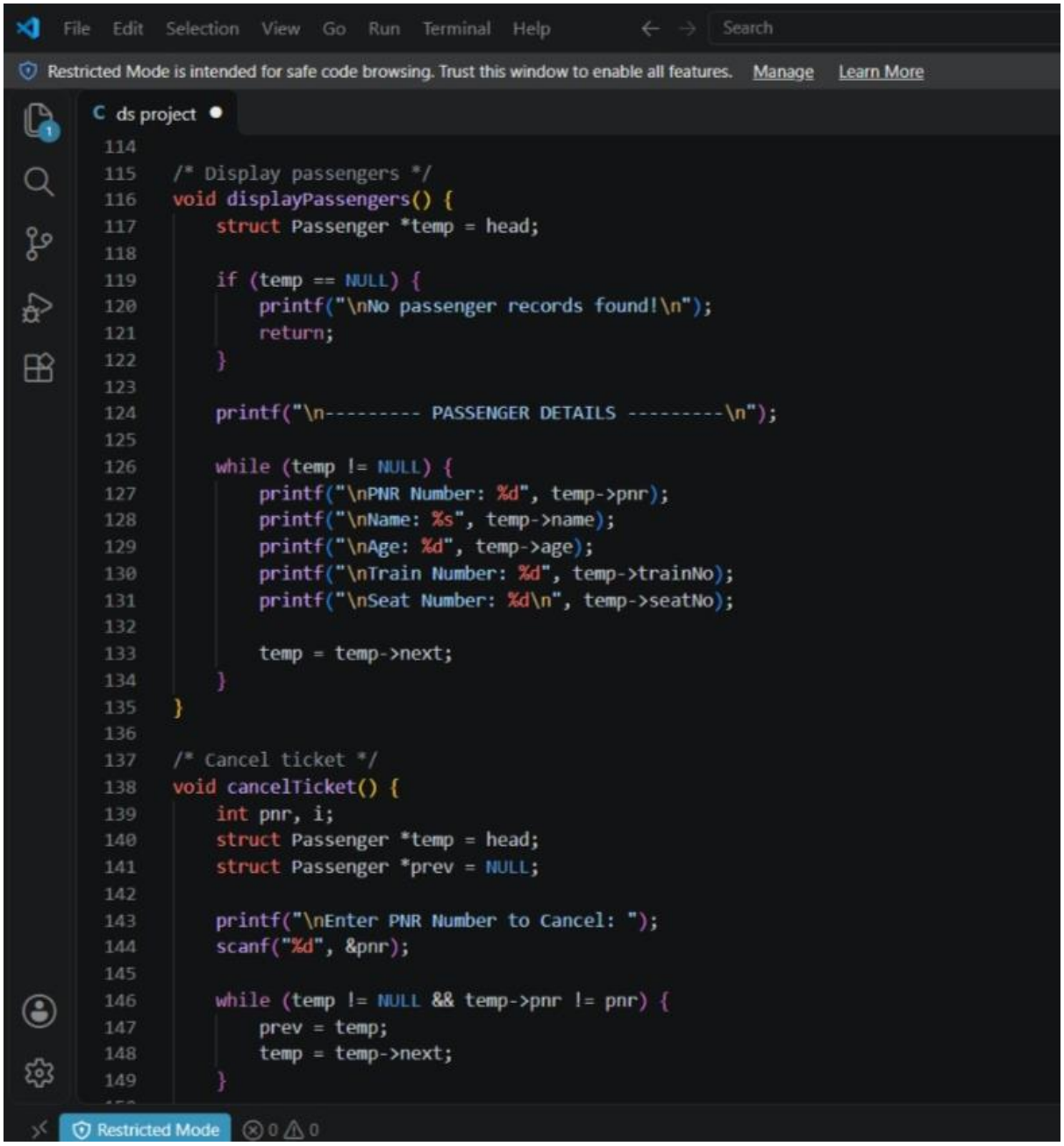

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ds project
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX_SEATS 10
struct Passenger {
int pnr;
char name[50];
int age;
int trainNo;
int seatNo;
struct Passenger *next;
};
struct Passenger *head = NULL;
int nextPNR = 1001;
struct Train {
int trainNo;
char trainName[50];
char source[30];
char destination[30];
int availableSeats;
};
struct Train trains[] = {
{101, "Chennai Express", "Chennai", "Bangalore", MAX_SEATS},
{102, "Pandian Express", "Chennai", "Madurai", MAX_SEATS},
{103, "Cheran Express", "Chennai", "Coimbatore", MAX_SEATS}
};
int trainCount = 3;
/* Display train details */
void displayTrains() {
Restricted Mode


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ds project
/* Display train details */
void displayTrains() {
int i;
printf("\n--------- TRAIN DETAILS ---------\n");
printf("Train No\tTrain Name\t\tSource\t\tDestination\tSeats\n");
for (i = 0; i < trainCount; i++) {
printf("%d\t\t%s\t%s\t%s\t%d\n",
trains[i].trainNo,
trains[i].trainName,
trains[i].source,
trains[i].destination,
trains[i].availableSeats);
}
}
/* Book ticket */
void bookTicket() {
int trainNo, age, i;
char name[50];
printf("\nEnter Passenger Name: ");
scanf(" %49[^\n]", name);
printf("Enter Age: ");
scanf("%d", &age);
if (age <= 0 || age > 120) {
printf("Invalid age!\n");
return;
}
displayTrains();
Restricted Mode
0 0




```
53  void bookTicket() {
87
88      struct Passenger *newPassenger =
89          (struct Passenger *)malloc(sizeof(struct Passenger));
90
91      if (newPassenger == NULL) {
92          printf("Memory allocation failed!\n");
93          return;
94      }
95
96      newPassenger->pnr = nextPNR++;
97      strcpy(newPassenger->name, name);
98      newPassenger->age = age;
99      newPassenger->trainNo = trainNo;
100     newPassenger->seatNo =
101         MAX_SEATS - trains[i].availableSeats + 1;
102
103     newPassenger->next = head;
104     head = newPassenger;
105
106     trains[i].availableSeats--;
107
108     printf("\nTicket Booked Successfully!\n");
109     printf("PNR Number: %d\n", newPassenger->pnr);
110     printf("Passenger: %s\n", newPassenger->name);
111     printf("Train: %s\n", trains[i].trainName);
112     printf("Seat Number: %d\n", newPassenger->seatNo);
113 }
114
115 /* Display passengers */
116 void displayPassengers() {
117     struct Passenger *temp = head;
118
119     if (temp == NULL) {
120         printf("\nNo passenger records found!\n");
121         return;
```




```c
/* Display passengers */
void displayPassengers() {
    struct Passenger *temp = head;

    if (temp == NULL) {
        printf("\nNo passenger records found!\n");
        return;
    }

    printf("\n--------- PASSENGER DETAILS ---------\n");

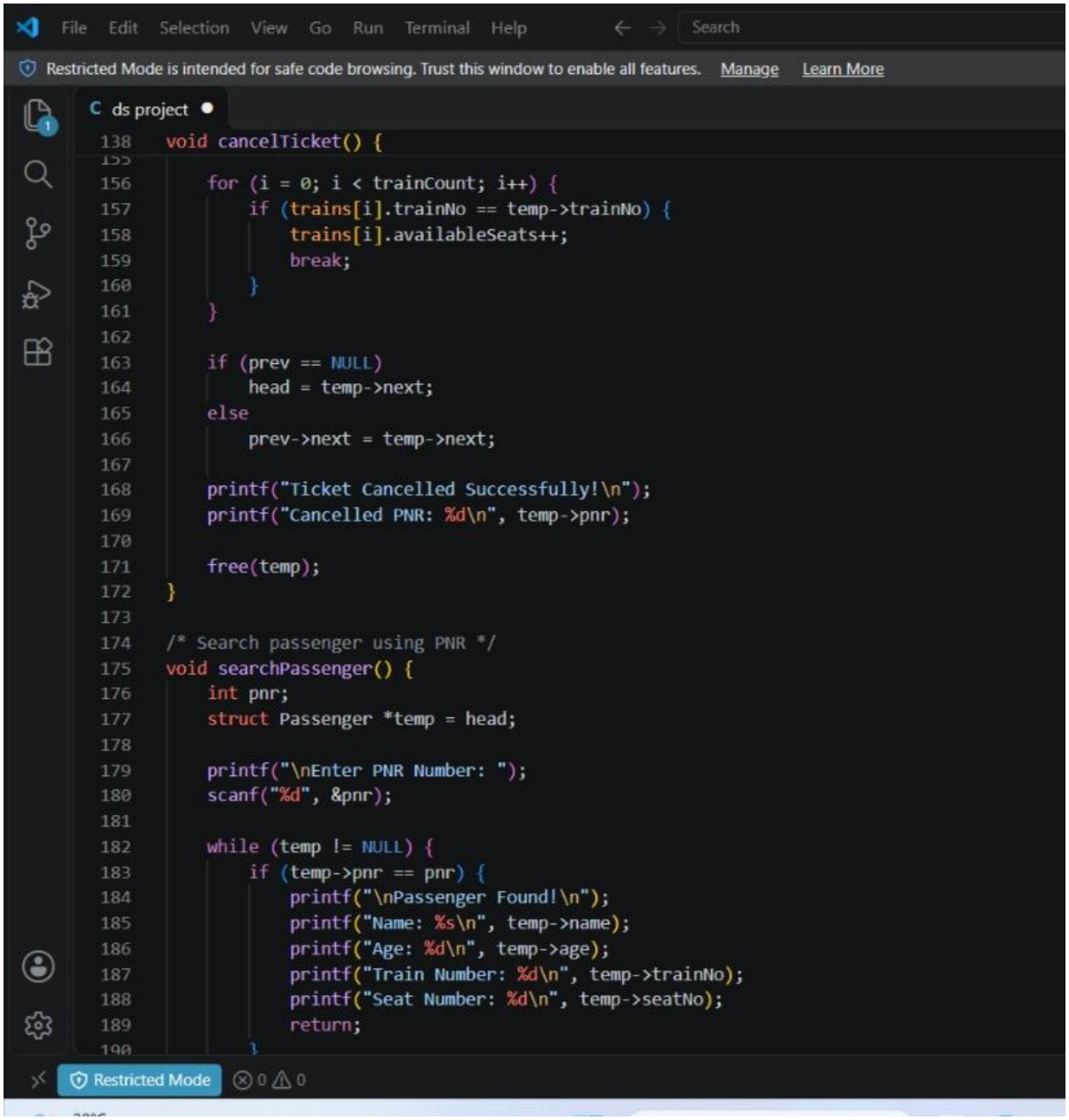
    while (temp != NULL) {
        printf("\nPNR Number: %d", temp->pnr);
        printf("\nName: %s", temp->name);
        printf("\nAge: %d", temp->age);
        printf("\nTrain Number: %d", temp->trainNo);
        printf("\nSeat Number: %d\n", temp->seatNo);

        temp = temp->next;
    }
}

/* Cancel ticket */
void cancelTicket() {
    int pnr, i;
    struct Passenger *temp = head;
    struct Passenger *prev = NULL;

    printf("\nEnter PNR Number to Cancel: ");
    scanf("%d", &pnr);

    while (temp != NULL && temp->pnr != pnr) {
        prev = temp;
        temp = temp->next;
    }
```




```c
void cancelTicket() {

    for (i = 0; i < trainCount; i++) {
        if (trains[i].trainNo == temp->trainNo) {
            trains[i].availableSeats++;
            break;
        }
    }

    if (prev == NULL)
        head = temp->next;
    else
        prev->next = temp->next;

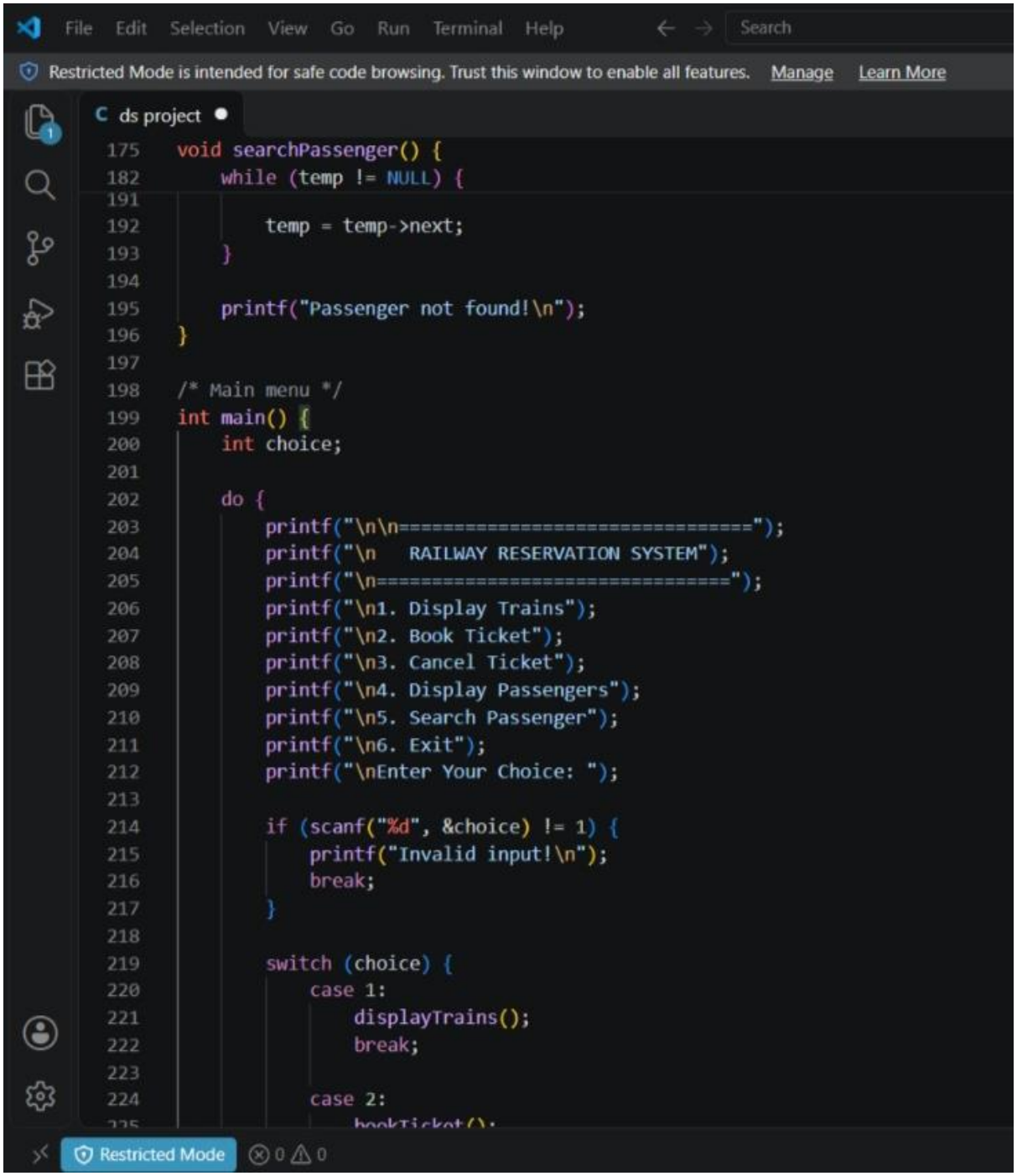
    printf("Ticket Cancelled Successfully!\n");
    printf("Cancelled PNR: %d\n", temp->pnr);

    free(temp);
}

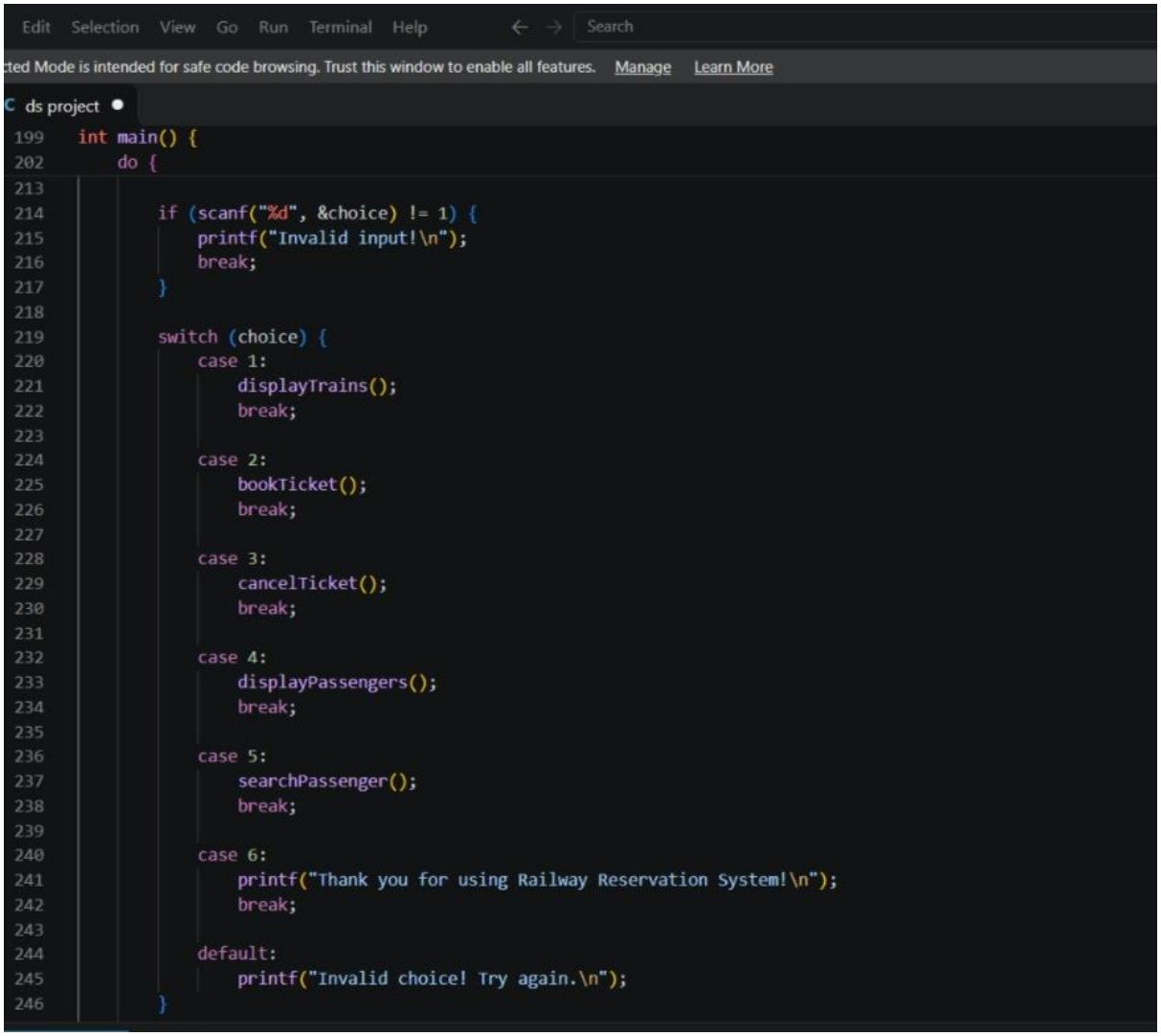
/* Search passenger using PNR */
void searchPassenger() {
    int pnr;
    struct Passenger *temp = head;

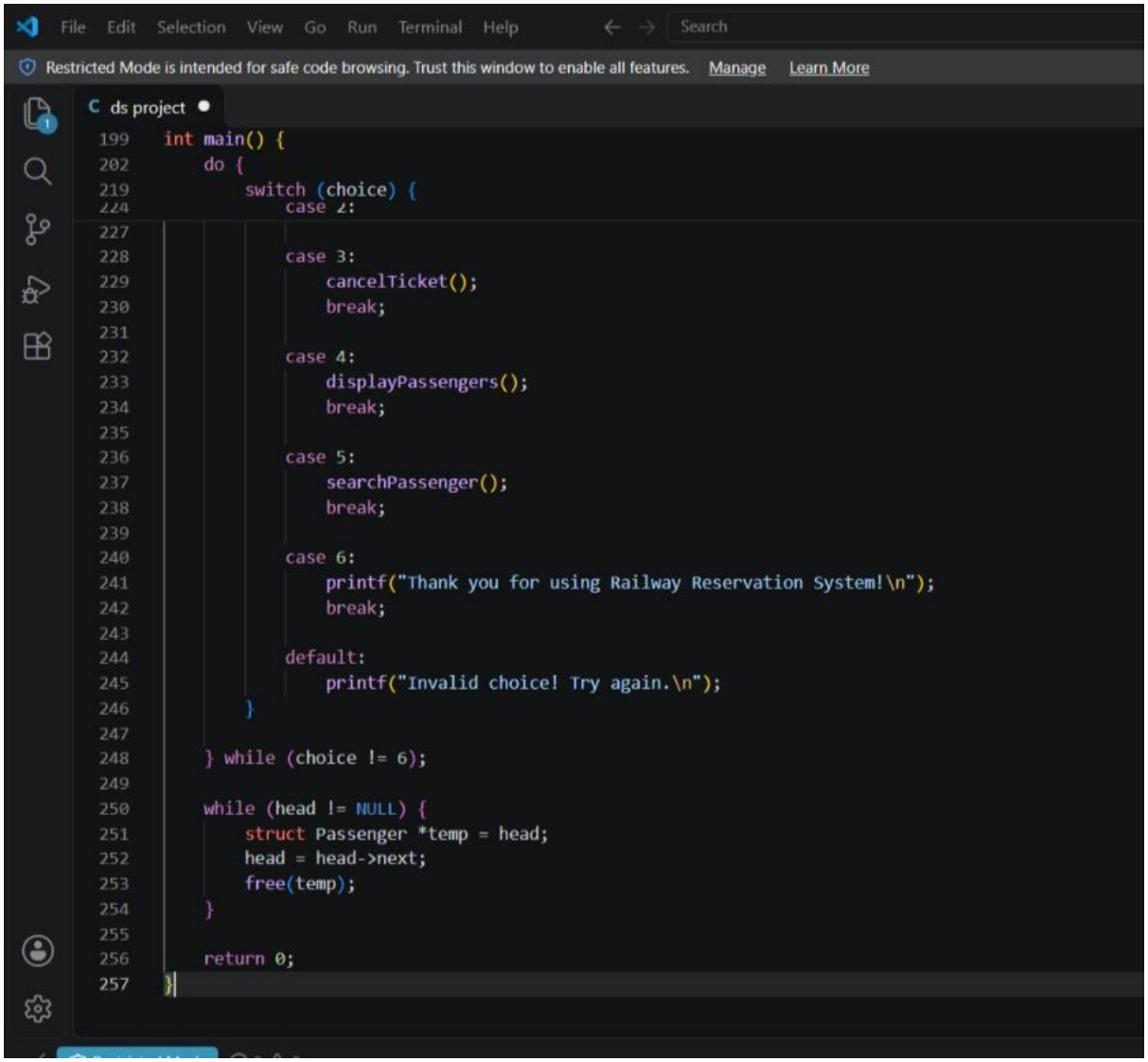
    printf("\nEnter PNR Number: ");
    scanf("%d", &pnr);

    while (temp != NULL) {
        if (temp->pnr == pnr) {
            printf("\nPassenger Found!\n");
            printf("Name: %s\n", temp->name);
            printf("Age: %d\n", temp->age);
            printf("Train Number: %d\n", temp->trainNo);
            printf("Seat Number: %d\n", temp->seatNo);
            return;
        }
```




```
175  void searchPassenger() {
182      while (temp != NULL) {
191
192          temp = temp->next;
193      }
194
195      printf("Passenger not found!\n");
196  }
197
198  /* Main menu */
199  int main() {
200      int choice;
201
202      do {
203          printf("\n\n==============================");
204          printf("\n   RAILWAY RESERVATION SYSTEM");
205          printf("\n==============================");
206          printf("\n1. Display Trains");
207          printf("\n2. Book Ticket");
208          printf("\n3. Cancel Ticket");
209          printf("\n4. Display Passengers");
210          printf("\n5. Search Passenger");
211          printf("\n6. Exit");
212          printf("\nEnter Your Choice: ");
213
214          if (scanf("%d", &choice) != 1) {
215              printf("Invalid input!\n");
216              break;
217          }
218
219          switch (choice) {
220              case 1:
221                  displayTrains();
222                  break;
223
224              case 2:
                    bookTicket();
```


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199 int main() {
202 do {
213
214 if (scanf("%d", &choice) != 1) {
215 printf("Invalid input!\n");
216 break;
217 }
218
219 switch (choice) {
220 case 1:
221 displayTrains();
222 break;
223
224 case 2:
225 bookTicket();
226 break;
227
228 case 3:
229 cancelTicket();
230 break;
231
232 case 4:
233 displayPassengers();
234 break;
235
236 case 5:
237 searchPassenger();
238 break;
239
240 case 6:
241 printf("Thank you for using Railway Reservation System!\n");
242 break;
243
244 default:
245 printf("Invalid choice! Try again.\n");
246 }


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int main() {
do {
switch (choice) {
case 2:
case 3:
cancelTicket();
break;
case 4:
displayPassengers();
break;
case 5:
searchPassenger();
break;
case 6:
printf("Thank you for using Railway Reservation System!\n");
break;
default:
printf("Invalid choice! Try again.\n");
}
} while (choice != 6);
while (head != NULL) {
struct Passenger *temp = head;
head = head->next;
free(temp);
}
return 0;
}

```
================================
   RAILWAY RESERVATION SYSTEM
================================
1. Display Trains
2. Book Ticket
3. Cancel Ticket
4. Display Passengers
5. Search Passenger
6. Exit
Enter Your Choice: 1

--------- TRAIN DETAILS ---------
Train No    Train Name      Source      Destination Seats
101     Chennai Express Chennai Bangalore   10
102     Pandian Express Chennai Madurai 10
103     Cheran Express  Chennai Coimbatore  10
```

```
Enter Passenger Name: j.k.jayasri
Enter Age: 18

--------- TRAIN DETAILS ---------
Train No    Train Name      Source      Destination Seats
101     Chennai Express Chennai Bangalore   10
102     Pandian Express Chennai Madurai 10
103     Cheran Express  Chennai Coimbatore  10

Enter Train Number: 101

Ticket Booked Successfully!
PNR Number: 1001
Passenger: j.k.jayasri
Train: Chennai Express
Seat Number: 1
```